

[RVHS/2024/Q2]

2 ASCII image

Name your Jupyter Notebook as:

Task_2_<class>_<index number>_<your name>.ipynb

For each of the sub-tasks, complete the codes in Jupyter Notebook and add a comment statement at the beginning of the code using the hash symbol #, to indicate the sub-task the program code belongs to, for example,

```
In [1]: #Task 2_1
        Program code
```

Output:

An ascii image is an image that is made up of ascii characters like the one below.

\$\\$ \backslash	\$\\$ \backslash	\$\\$ \\$\\$ \\$\\$ \backslash	\$\\$ \backslash	\$\\$ \backslash
\\$ \\$ \backslash	\$\\$	\\$\\$	\$\\$ \backslash	\$\\$
\\$ \\$ \backslash	\$\\$ /	\$\\$ /	\$\\$	\$\\$
\\$ \\$ \\$ \\$	/	\$\\$	\$\\$	\$\\$
\\$ \\$	/	\$\\$	\$\\$	\$\\$
\$\\$		\$\\$	\$\\$	\$\\$
\$\\$		\$\\$ \\$\\$ \\$\\$	\\$ \\$ \\$ \\$ \\$ \\$	
\		\	/	\

In this task, you will read a list of ascii code and recreate the ascii image by writing it into an output text file.

Task 2.1

Write a function `task 2 1(filename)` that:

- takes a string value, `filename`
- converts each decimal ascii code in the file into its actual ascii character
- returns a list of ascii characters in the same order as it is in the file. [2]

For example, if `filename` contains the following.

45, 32, 40, 40, 99, 32, 45, 45, 45, 45, 45, 45, 45, 32, 41,
32, 32, 32, 45, 45, 45, 45, 45, 45, 32, 92, 32, 40, 32, 32,
32, 32, 32, 34, 32, 32, 32, 32, 32, 32, 47, 46, 34, 41, 40,
41, 32, 41, 46

[illegible]

Test your function using the following call:

- ```
• task 2 1("inputA.txt") [1]
```

[RVHS/2024/Q2 Cont']

## Task 2.2

Write a function `task_2_2(char_1st)` that:

- takes a list of ascii characters, `char_lst`
  - compute the frequency of each ascii character appeared in `char_lst`
  - sorts the ascii characters according to their frequencies using insertion sort
  - displays the ascii character and its frequency in the **same** format as shown in the example below.
- [6]

For example, `char_1st` contains the following.

[illegible]

```
>>> task_2_2(char_lst)
```

| Char  | Freq |
|-------|------|
| ' '   | 20   |
| ' - ' | 14   |
| ' ) ' | 4    |
| ' ( ' | 4    |
| ' . ' | 2    |
| ' " ' | 2    |
| ' / ' | 1    |
| ' \ ' | 1    |
| ' c ' | 1    |

Test your function using the following call:

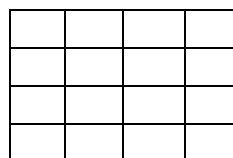
- `task_2_2(task_2_1("inputA.txt"))` [3]

## Task 2.3

For this task, you are required to insert a list of characters spirally into a 2-dimension square matrix. Consider a 16 characters list as follows:

```
#abcdefghijklmn@
```

Starting from the top left corner of the 2D square matrix, insertion of characters starts downward, then rightward, upward and leftward. The process repeats until the whole 2D square matrix is filled up and ended at the center. Hence, the 16 characters list will be inserted as follows:



## [RVHS/2024/Q2 Cont']

Write a function `task_2_3(char_lst)` that:

- takes a list of ascii characters, `char_lst`
- inserts each character of `char_lst` in a spiral way into a 2D list of size  $n \times n$  where  $n$  is the square root of the length of `char_lst`.
- returns the 2D list.

For example, `charset` contains the following.

[illegible]

```
>>> task 2 3(char lst)
```

[illegible]

Run the following code if you would like to visualise the image.

```
result = task_2_3(task_2_1("example.txt"))
for row in result:
 for char in row:
 print(char, end = "")
 print()
```

The following image should be displayed.

\_\_\_\_\_

$$\begin{array}{l} (\backslash /) \\ ( \quad . \quad . ) \\ \subset ( " ) ( " ) \end{array}$$

\_\_\_\_\_

### Task 2.4

Write a function `task 2 4(input file, outputfile)` that:

- takes two string values `input_file` and `outputfile`
- processes the content of `input_file` using `task_2_1()` and `task_2_3()`
- writes the result from `task_2_3()` in `outputfile` with each row of the 2D array occupies a line and each character in the line adjacent to each other. [3]

Test your function using the following call:

- `task_2_4("example.txt", "output.txt")`
- `task_2_4("inputA.txt", "outputA.txt")`
- `task_2_4("inputB.txt", "outputB.txt")`

[1]